applied science—using knowledge to develop practical solutions relevant to contemporary environmental issues. American environmentalist Jane Lubchenco led the SBI, and paved the way for the ESA (and others) to promote useful ecological knowledge as scientists raced to combat environmental degradation.

“My goal [is] to change the culture so it becomes unthinkable to not use evidence in conservation.”

William J. Sutherland

Prioritizing the planet

The scientists of the SBI set out a new path for the field of ecology, and determined which research areas would be the most important in the years to come. They sought to prioritize three fields of research: global change, biological diversity, and sustainable ecological systems. Studies of global change look at the atmosphere, climate, soil and water (including changes due to pollution), and patterns of land- and water-use. Research into biological diversity focuses on the conservation of endangered species and the study of natural and human-made changes in genetic and habitat diversity. Finally, studies of sustainable ecological systems analyze the interactions between humans and ecological processes in order for scientists to find solutions to the stresses they detect in ecosystems.

The SBI stressed the need for funding for such research, and also highlighted the importance of sharing findings with those outside the scientific community. It set out a process for applied ecological research that included not only acquiring new knowledge, but communicating it and helping incorporate it into real-world policy changes.